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Deep-Learning Approach for An Analysis of Real-estate Prices and Transactions

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ABSTRACT Research on the study of houses, condominiums and buildings in Taiwan's metropolitan areas continues to be an important area of research. In real estate forecasting and analysis, methods such as statistical analysis and questionnaire data collection are widely used. However, when multidimensional data is considered, these methods are time-consuming and inadequate. This study aimed to build a real estate forecasting model that can adapt to a changing environment. Data were collected from public government databases, the collected data were standardized for accurate clustering, an appropriate data clustering algorithm was applied to the standardized data, and cross-statistical analysis was performed to verify the adopted algorithm. We used a deep learning based on autoencoder algorithm to increase the accuracy of the clustering analysis. A double-bottom map particle swarm optimization (DBM-PSO) clustering algorithm was then used to determine the optimal clustering solution. Cluster analysis and deep learning were conducted on data collected from public websites to understand the factors that led to the sustained increase in housing prices in Taiwan over the past decade. The results of this study indicate that three key factors—the number of real estate transactions, the average unit price of real estate transactions, and the building material and construction index—significantly affected real estate prices in Taiwan. Our results could help researchers and governments to focus on specific aspects of real estate development without being influenced by other related factors. In addition, the relationships between real estate trends and the aforementioned three key factors were determined to obtain valuable information that can enable the Taiwanese government to regulate the property market and prevent excessive growth. The framework proposed in this paper allows researchers and governments to focus on specific aspects of real estate development without being influenced by other related factors, and provides a new mechanism for approaching real estate-related forecasting.

INDEX TERMS Machine learning; real estate; particle swarm optimization algorithm; economy

I. INTRODUCTION

The expression "own the land, own the wealth" is highly common in Taiwanese culture, and most families in Taiwan aim to buy a home. Land and permanent structures—such as houses, apartments, buildings, and fences—are considered to be real estate [1]. Real estate symbolizes security and stability and serves as a form of shelter. It has become a popular investment and a reliable hedge against inflation on a global scale. The state of the economy affects housing demand, with property values and rental rates rising with increases in inflation.

Open databases enable researchers to access established data for analysis. The Federal Reserve Economic Data database (<https://fred.stlouisfed.org>) of the United States contains information on various housing indicators, including housing inventory, housing starts, home sales, the housing affordability index, the housing price index, housing transactions, the mortgage-lending index, commercial property prices, interest rates, the producer price index, and foreclosed properties. Real estate data are complex and unpredictable, and most of the real estate literature focuses on the trend in housing prices [2] and housing price forecasting [3]. Unit price, initial property registration, and number of

rooms are essential aspects in real estate research. Unexpected uncertainties such as pandemic outbreaks, market volatility, and economic crises make real estate data unpredictable. In recent years, advanced machine learning algorithms have been developed to account for such uncertainties in the long-term forecasts [4].

Voith et al. developed a model that decomposes the variation in office vacancy rates into market-specific, time-specific, and random factors [5]. An understanding of real estate trends is crucial for the prediction of future prices [6]. Drawing on economics, sociology, psychology, and methodological and epistemological philosophy, Mooya et al. raised three fundamental questions about the measurement and magnitude of real estate prices [7]. Yang stated that since the recovery of Taiwanese real estate, its historical changes and development trends have been affected differently by Taiwan's economic development, population growth, major natural disasters, and related government policies. In the short and long term, macroeconomic variables such as income, permissible floor area, and housing loan balance can strongly affect the housing price index. In addition to objective conditions, some psychological factors influence housing prices.

Over the past few decades, housing prices in Taiwan have risen steadily. Home buyers have continued to purchase homes despite believing that housing prices are unreasonable. Monthly sectoral stock market indices provide valuable information for real estate analysis. Trends in the sociodemographic factors that influence the spatial structure of the real estate market might be useful for segmenting the housing market in accordance with neighborhood socioeconomic status [8]. Since the Global Financial Crisis in 2007–2008, the real estate market has received considerable research attention [9]. The relationships of real estate with socioeconomic and sociodemographic factors have become a critical field of study. Moreover, the relationship between the stock index and exchange rate has attracted considerable scholarly interest [10]. The stock market serves as a proxy for the economic performance of various countries, and daily stock market indices provide a more accurate assessment of local economic conditions than do other factors based on less timely data [11, 12]. Predicting real estate trends is essential for guiding local government policies, formulating real estate products, balancing society's real estate needs, and improving the competitiveness of construction companies.

The real estate market is characterized by collective behaviors resulting from its complex system. Data mining through clustering is a practical approach for analyzing diverse real estate data. The separation of data into different clusters in accordance with patterns without predefined labels provides insight into the structure of data and enables the identification of meaningful data patterns. Various clustering methods have been applied to real-world real estate problems and have produced successful results [13, 14]. For example, *k*-means clustering was used by Kan-Kilinc [15] to analyze real estate prices in Istanbul, Turkey. Liao's data mining results indicated the

existence of strong associations between Taiwan's stock market indices and the electronics, finance and insurance, semiconductor, and TAIEX indices [16]. Clustering is an essential statistical analysis tool and is becoming increasingly important in data mining for identifying structures in large data sets [17].

Historical fluctuations in real estate demand can be used to forecast future trends of the real estate market [18]. Government policies and regulations can be informed by monitoring the development of the national real estate industry and the relationship between the supply and demand of real estate. Real estate companies must forecast the real estate market accurately when making investment decisions regarding real-estate-related businesses. An understanding of homebuyer demand and housing inventory is critical for determining market demand. Forecasting can provide valuable insight for developers wishing to adjust their business strategies and minimize their risk. Companies can use real estate market forecasts to develop plans that are in line with the market demand, thus meeting market requirements while reducing business risks.

Machine learning algorithms focusing on the analysis of factors such as real estate prices and investment opportunities have found application in real estate studies. Several estimation techniques are used, including regression analysis [19] and machine learning methods like neural networks [20], decision trees [21], random forests [22], and ensemble methods [23]. Predicting house prices, understanding the impact of house characteristics on prices, and developing hedonic price indices are the three main objectives of these approaches. The application of machine learning methods to the estimation of property values within specific locations is challenging. Specifically, clustering represents a crucial yet complicated task in fields such as machine learning and pattern recognition. However, the effective handling of large-scale, semantically complex, high-dimensional data is still a challenge [24]. This challenge arises from two main situations: the need to represent data in lower dimensions and ensuring clustering efficiency. One of the most promising ways to address these challenges is the application of autoencoders and their variants to unsupervised learning tasks. Autoencoders are a useful tool for the learning of node representations in graphs. They introduce a regularization term that induces a contraction of the mapping between the input space and the feature space around training samples [25]. By incorporating constraints on the contraction ratio and the Jacobian matrix into the loss function, autoencoders suppress perturbations in all directions, which results in robust extracted features. Furthermore, *k*-means clustering becomes popular after autoencoder [26–29]. However, it's important to note that the *k*-means approach has limitations in its implementation for clustering analysis in related studies [17, 30, 31].

Governments aim to regulate housing prices to prevent real estate speculation and ensure that housing access is equitable. Governments can set appropriate interest rates on loans and implement other policy measures to stabilize housing prices.

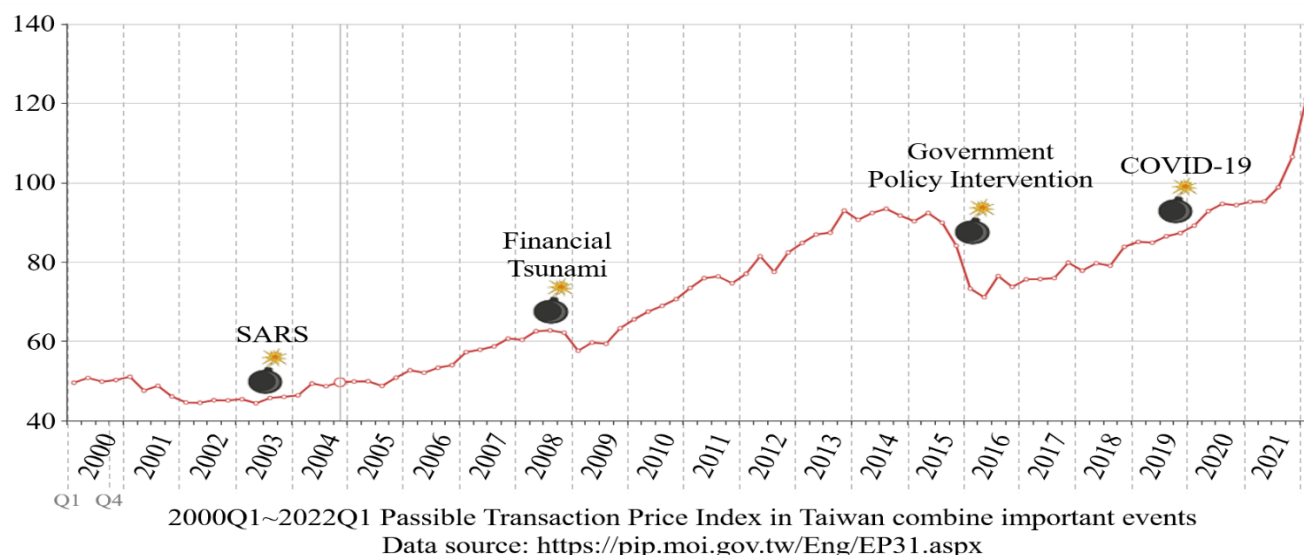


Figure 1. Transaction price index in Taiwan combines actual events (2021 base period = 100).

According to the website of the Taiwanese Ministry of the Interior (<https://pip.moi.gov.tw/Eng/EP31.aspx>), housing prices in Taiwan increased between 2000 and 2022. The trend in Taiwan's transaction index between Q1 in 2000 and Q2 in 2022 is displayed in Figure 1. Government policies influence the behavior of Taiwan's real estate market. Therefore, the need for government intervention to maintain stability is evident from Taiwan's real estate history. The development of property prices during key events in Taiwan is illustrated in Figure 1. Real estate has become a crucial research area because of its importance to the public. Taiwan is a small country, and the Taiwan real estate market is one of the relatively closed markets in the world. Different perspectives on real estate, which are shaped by personal and social values, are held by residents of the western and eastern coasts of Taiwan [32]. The economy and income of Taiwan are more developed than Southeast Asia's, and foreign investment in Taiwan has been relatively high. In this study, our research question is to examine on houses, apartments, and buildings in Taiwan's metropolitan areas. Therefore, there are indeed many factors that have an impact on the real estate forecast. To more thoroughly explore and challenge the accuracy of Taiwan's real estate forecast, we collect all the relevant data in Taiwan. The data collection groups can be divided into six groups:

1. The house registration indexes: all major real estate registration indexes, including total price, unit price, area of house transactions, and house age. The respective indicators have important correlations with the house registration indices that can be identified.
2. The real estate financial indicators: Mortgage Interest Rate, M1b, M2, Rental Index, and Consumer Price Index affect the house transaction. The correlation can be used as a real estate trend forecast by cross-discussing real estate transaction data and financial indicators.

3. The real estate transaction information: The number of first-time real estate registration, the average floor area of first-time registration, the average unit price of house transactions, and the number of first-time real estate registration record the information of house transactions. With the real estate transaction information (new house/old house/square number/unit price/total price), the transaction trend of housing layout/square number and unit price can be determined.
4. The Demographic Population Indices: Taiwan's demographic structure is divided into three groups. In this study, Taiwan's population is divided into three groups to count the population (children 0-14 years old, middle-aged 15-64 years old, and elderly over 65 years old). The impact of the population structure on real estate transaction trends can be studied.
5. The classified stock market indexes: Taiwan is an economically developed country, and the stock market is an economic indicator of a country, which also affects the real estate transaction market. The important thing to explore the classified stock price is to examine the industry's impact on real estate.
6. The Construction Cost Indices: Construction costs have always affected housing prices and transaction volume. We can cross-reference the impact on construction costs, industrial economy, and real estate transactions by exploring the impact of various construction costs on real estate trends.

For novelty, this study is firstly to investigate the factors driving the sustained increase in Taiwan's housing prices over the past decade by collecting and analyzing open-source real estate data. Various techniques—including data mining, standardization, and classification—were used to process the collected data. Taiwan Stock Exchange Corporation (TWSE) data on real estate in Taiwan were subjected to segmentation.

Moreover, an algorithmic framework involving cluster analysis and deep learning was designed to identify the key characteristics that affect real estate prices. An autoencoder algorithm was then employed to increase the accuracy of the clustering analysis, and a double-bottom map particle swarm optimization (DBM-PSO) clustering algorithm was used to determine the optimal clustering solution. In addition, a cohort analysis method was employed to obtain insights into the real estate market in Taiwan. In this analysis, the correlations of various factors—such as sociodemographic factors, socioeconomic data, and stock market indices—with the average unit price, number of real estate registrations, and floor area of houses was investigated. In brief, our main contributions are as follows:

1. We build a framework based clustering and deep learning to identify features of real estate. An autoencoder algorithm was employed to increase the clustering accuracy, and DBM-PSO clustering algorithm used to determine the optimal clustering solution.
2. The results of this study could serve as a reference for governments and real estate investors in developing policies, regulations, and market strategies as well as forecasting trends accurately to maintain the relationships between real estate characteristics.

The rest of this paper is organized as follows. Section 2 presents a case study and describes the clustering of data by using the autoencoder method. Section 3 presents the results of data clustering, details the advantages of the adopted data clustering approach, and describes the effects of marketing on the real estate market in terms of stock market indices. Finally, section 4 provides the conclusion of this study.

II. METHODS

The proposed clustering model is outlined in this section. First, a four-layer autoencoder network minimizes the reconstruction error and the distance between the data points and their respective clusters in the code layer. Second, the data representation in the data layer is improved through nonlinear mapping. This process consists of four main steps: (1) standardizing the data for various features and attributes, (2) reducing the multidimensional data to a low-dimensional data through dimensionality reduction based on autoencoder approach, (3) performing clustering in the low-dimensional space, and (4) analyzing the obtained clusters related to real estate markets. A detailed description of the adopted autoencoder network is provided in the following sections, after which its application in clustering is clarified.

A. Data standardization

Data were obtained from various sources, included multiple types of information, and described various characteristics and attributes. Therefore, we used mean normalization [33] to standardize the ranges of the independent variables (x). The original variable file can be found in the Supplementary Files (98index.xlsx).

$$x' = \frac{x - \mu}{\max(x) - \min(x)} \quad (1)$$

where μ is the mean value of the independent variable. The original data and the descriptive statistics of the collected variables can be found in the Supplementary Files (98index.xlsx).

B. Reducing the high-dimensional models using Auto-encoders

An autoencoder is a type of neural network that aims to learn a hidden representation of its input to then reconstruct the input [25]. Consider a single-layer autoencoder network that comprises an encoder and a decoder. The encoder uses a nonlinear mapping function $f(x)$ to map an input x_i to the input's hidden representation h_i . The activation function used in this study is the sigmoid function, which is defined as follows:

$$h_i = f(x_i) = \frac{1}{1 + \exp(-(W^{(1)}x_i + b^{(1)}))} \quad (2)$$

where $W^{(1)}$ is the encoding weight matrix and $b^{(1)}$ is the bias vector of the encoding. If the activation function is linear, the operation of the one-layer autoencoding network is equivalent to principal component analysis. The decoder aims to reconstruct the input x_i from its hidden representation h_i by using the decoder function g , which is expressed as follows:

$$x'_i = g(h_i) = \frac{1}{1 + \exp(-(W^{(2)}h_i + b^{(2)}))} \quad (3)$$

where $W^{(2)}$ is the decoding weight matrix and $b^{(2)}$ is the bias vector of the decoding. The condition $W^{(2)} = W^{(1)T}$ can act as a regularizer and reduce the number of parameters to optimize [34]. The optimal parameters of W and b have to be defined during the coding process. The goal of an autoencoder is to minimize the error of the reconstruction process. The cost function shown in equation (4) can be solved to obtain these two parameters by minimizing $\text{cost}(\theta)$, since it is directly related to the parameters W and b .

$$\text{cost}(\theta) = \frac{1}{N} \sum_{i=1}^N \|x_i - x'_i\|^2 \quad (4)$$

where $\theta = \{W^{(1)}, W^{(2)}, b^{(1)}, b^{(2)}\}$, x'_i is calculated as Equation (3). A deep autoencoder network can be formed by stacking several one-layer autoencoders. Here, the hidden representation of a previous autoencoder is used as input for the next autoencoder. However, because the gradient descent algorithm used for training is sensitive to initial weights [35], deep autoencoders can be difficult to train. To overcome this problem, a pre-training algorithm is used, which learns more robust features than the gradient descent algorithm before fine-tuning the entire model. Restricted Boltzmann machines [35] and denoising autoencoders [36] are used to construct a model

that can reconstruct inputs from their distorted versions. Denoising autoencoders can be considered stochastic versions of autoencoders. The difference between a denoising autoencoder and a conventional autoencoder is that a denoising autoencoder performs a stochastic corruption process. This process randomly sets some inputs to 0. A denoiser autoencoder is trained to reconstruct the undistorted version of the input from the distorted version. Thus, the loss function in Eq. 4 is used in this autoencoder. As a result, the learning process attempts to cancel out the stochastic corruption process by capturing the statistical relationships between the inputs. After all layers have been pre-trained (i.e., without the corruption process), the autoencoder is fine-tuned. We define an autoencoder with two dense layers for encoder, which compresses the data set into a 36-dimensional latent vector, and two dense layers for decoder, which reconstructs the original data from the latent space. A four-layer autoencoder network (Fig. 2) minimizes the reconstruction error and the distance between the data points and their respective clusters in the code layer. We used 75% and 50% feature reduction for to detect the 25% features in feature generating layer. In our data set, we have 144 features as input vector. Therefore, an autoencoder with four layers in encoder and decoder have the following numbers of neurons: encoder₁ = 108, encoder₂ = 72 and decoder₁ = 72, decoder₂ = 108. The input data is time-indexed independent variable.

C. Clustering the low-dimensional space

Particle Swarm Optimization

Kennedy and Eberhart [37] introduced the particle swarm optimization (PSO) algorithm. PSO is an efficient learning algorithm based on evolutionary computation that mimics the movements of flocks of birds to simulate social behavior [38]. Each candidate solution is treated as a particle in a swarm, and each particle's velocity can be adjusted in accordance with its experience and the information that it exchanges with other particles in the swarm. The PSO algorithm effectively finds optimal solutions in complex search spaces through interactions between individuals in a population of particles.

The PSO process can be divided into several steps. First, the search space is defined as the feasible region of the problem and is represented as the set of all feasible solutions. The swarm consists of M particles, which are expressed as $S = (X_1, X_2, \dots, X_M)$. Second, each particle is initialized with a random solution and is represented as a point in a D -dimensional space, where $X_i = (x_{i1}, x_{i2}, \dots, x_{iD})$ and $x \in (X_{\min}, X_{\max})^D$. A particle's velocity is represented as $v_i = \{1, 2, \dots, v_{iD}\}$, where $v \in (V_{\min}, V_{\max})^D$. Third, each particle finds its best position by comparing its current position's fitness with that of its previous best position, which is represented as $pbest_i = \{pbest_{i,1}, pbest_{i,2}, \dots, pbest_{i,D}\}$. Fourth, the best position, which is represented as $gbest = \{gbest_1, gbest_2, \dots, gbest_D\}$, is found among all the particles in the population. Finally, the PSO algorithm updates the position and velocity of each particle in

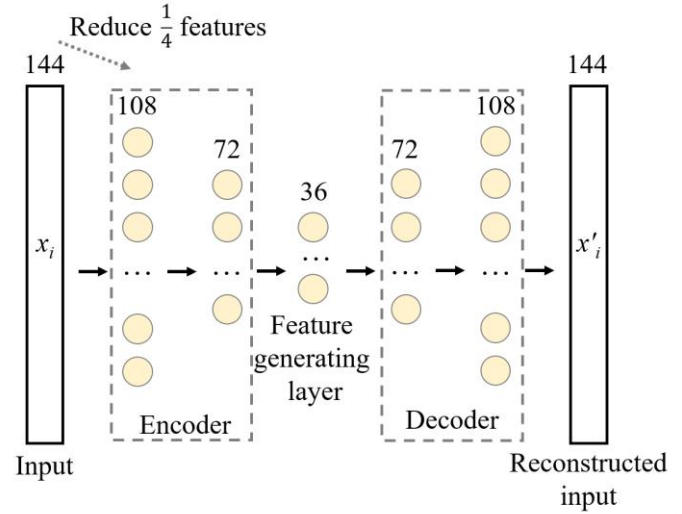


Figure 2. An architecture of autoencoder.

each generation with $pbest$ and $gbest$ to find the optimal solution. The equations for the update are as follows:

$$v_{i,d}^{t+1} = w \times v_{i,d}^t + c_1 \times r_1 \times (pbest_{i,d} - x_{i,d}^t) + c_2 \times r_2 \times (gbest_d - x_{i,d}^t) \quad (5)$$

$$x_{i,d}^{t+1} = x_{i,d}^t + v_{i,d}^{t+1} \quad (6)$$

The velocity of the particles is influenced by two constants (c_1 and c_2) and two random numbers (r_1 and r_2). The number of elapsed iterations is represented by t . The updated and current values of the velocity of the i th particle in the d th dimension are denoted $v_{i,d}^{t+1}$ and $v_{i,d}^t$, respectively. Moreover, the updated and current values of the position of the i th particle in the d th dimension are denoted $x_{i,d}^{t+1}$ and $x_{i,d}^t$, respectively. The inertia weight w decreases linearly from 0.9 to 0.4 as the number of iterations increases [39]. The inertia weight w can be expressed as follows:

$$w = (0.9 - 0.4) \times \frac{iteration_{max} - t}{iteration_{max}} + 0.4 \quad (7)$$

The terms w_{max} and w_{min} , which represent the upper and lower limits of w , respectively, are set to 0.9 and 0.4, respectively [40]. The terms $iteration_i$ and $iterations_{max}$ represent the current number of iterations and the total number of iterations, respectively. The velocity vector indicates how the particle moves to reach the global best position at a given instant. However, a particle's exploration of the solution space can be limited by velocity values that are excessively low or high. The particle's velocity and position are thus limited to within $[V_{\min}, V_{\max}]^D$ and $[X_{\min}, X_{\max}]^D$, respectively. These ranges define the maximum step size that the particle can use in the search space.

Double-Bottom Map Particle Swarm Optimization

DBM-PSO is a variant of PSO that uses a double-bottom map (DBM) in its updating function [41, 42]. A DBM is a chaotic function that prevents the population becoming trapped in a local optimum, and it balances exploitation and exploration in PSO search by generating random values in three regions: regions with values of 0.0, 0.5, and 1.0. In the PSO updating function, r_1 is related to updating the particle in accordance with $pbest$, and r_2 is related to updating the particle in accordance with $gbest$. In DBM-PSO, the DBM function generates two sequences, namely $DBMr_1$ and $DBMr_2$, which replace r_1 and r_2 , respectively, to balance global exploration and local search. The sequences $DBMr_1$ and $DBMr_2$ are expressed as follows:

$$DBMr_{i,d}^{t+1} = \frac{[\sin(4\pi DBMr_{i,d}^t) + 1]}{2} \quad (8)$$

where $DBMr_{i,d}^{t+1}$ is the DBMr sequence in the d th element of the i th particle in the current iteration t . The update function of DBM-PSO is expressed as follows:

$$\begin{aligned} v_{i,d}^{t+1} = & w \times v_{i,d}^t + c_1 \times DBMr_{1,i,d}^t \\ & \times (pbest_{i,d} - x_{i,d}^t) \\ & + c_2 \times DBMr_{2,i,d}^t \\ & \times (gbest_d - x_{i,d}^t) \end{aligned} \quad (9)$$

DBM-PSO Algorithm for Clustering

After the fine-tuning of the stacked autoencoder, it has learned the parameters θ . Subsequently, low-dimensional codes can be extracted from the code layer, and the DBM-PSO algorithm is used to cluster these codes. Algorithm 1 presents the pseudocode of the DBM-PSO clustering algorithm. The particle encoding, fitness evaluation, and $pbest$ and $gbest$ updates are defined for a specific clustering problem. The DBM-PSO clustering process involves the following steps:

Particle encoding: The position vector $\{x_{i,1}, x_{i,2}, \dots, x_{i,D}\}$ and velocity $\{v_{i,1}, v_{i,2}, \dots, v_{i,D}\}$ of a particle are randomly generated in the search space. The dimensions of the particle are represented by $K \times F$, where K is the total number of clusters and F is the total number of features in the data set.

Fitness evaluation: The fitness value indicates the sum of the intracluster distances within all clusters, and a lower fitness value indicates a lower error rate. The fitness value is calculated as follows:

$$\begin{aligned} \text{fitness} = & \sum D(x_j, z_i), i = 1, 2, \dots, K, j \\ & = 1, 2, \dots, L, \end{aligned} \quad (10)$$

where z_i is the i th cluster's centroid position, x_j is the j th sample position, K is the total number of clusters, and L is the total number of samples in the data set. The vector of the cluster center x_j is defined as follows:

$$z_i = \frac{1}{n_i} \sum_{x_j \in C_i} x_j \quad (11)$$

The matrix $x_j \in (C_1, C_2, \dots, C_i, \dots, C_K)$ represents the i th center of the K clusters. The Euclidean distance between x_j and z_i is calculated as follows:

$$D(x_j, z_i) = \sqrt{\sum_{d=1}^D (x_{j,d} - z_{i,d})^2} \quad (12)$$

The samples in the data set are assigned to the cluster with the shortest distance from them.

Updates of $pbest$ and $gbest$: The fitness value of a particle represents the sum of the intracluster distances within all clusters and is compared with the $pbest$ fitness value of the particle. If the particle's fitness is less than the $pbest$ fitness,

Algorithm 1. The steps of the DBM-PSO algorithm for clustering.

- 01 Initialize the particle swarm with M particles and their positions $X_i = (x_{i1}, x_{i2}, \dots, x_{iD})$ and velocities $v_i = \{1, 2, \dots, v_{i,D}\}$ in the D -dimensional search space.
- 02 Initialize the $pbest_i = \{pbest_{i,1}, pbest_{i,2}, \dots, pbest_{i,D}\}$ of each particle with its current position.
- 03 Initialize the $gbest = \{gbest_1, gbest_2, \dots, gbest_D\}$ of the particle swarm as the position of the particle with the best fitness.
- 04 While (the stop criterion is not satisfied)
- 05 For each particle (1 to M)
- 06 Calculate the fitness value (Eq. 10) for the particle by the evaluation function.
- 07 Update the $pbest$ position of the particle if the current fitness value is better than the personal best fitness value.
- 08 Update the $gbest$ position of the swarm if the current fitness value is better than the global best fitness value.
- 09 Update the particle's velocity using the velocity update equation (9), which includes the current velocity, $pbest$ difference from the current position, and $gbest$ difference.
- 10 Update the particle's position using the position update equation (6), which includes the current position and updated velocity.
- 11 End For
- 12 End While
- 13 Return to the $gbest$ position as the optimal solution.

the *pbest* fitness and position values are updated with the fitness and position of the particle, respectively. If the *pbest* fitness is greater than the *gbest* fitness, the *gbest* fitness and position values are updated with the *pbest* fitness and position values, respectively; otherwise, *gbest* and position remain unchanged.

Parameter settings: For DBM-PSO clustering control parameters, the maximum number of iterations is set to 100, and the total number of particles is set to 50. Coefficients c_1 and c_2 are set to 2. The maximum velocity ($V_{\max}$) is determined as the difference between the maximum and minimum values of X (i.e., $X_{\max} - X_{\min}$), and the minimum velocity ($V_{\min}$) is set as the negative of this difference [i.e., $-(X_{\max} - X_{\min})$] [37].

D. Dataset

We aimed to examine the effects of various factors on real estate trends in this study. We collected data from various sources, including the TWSE (for stock market sector indices), the Ministry of the Interior of Taiwan, the Ministry of Finance of Taiwan, Taiwanese national statistics, and public government data (for social, economic, and sociodemographic indicators). From the collected data, we selected 98 meaningful variables, including age group (0–14, 15–65, and >65 years), property transfer information (number of transfers, average transfer area, average unit price, number of houses constructed, and average building area), socioeconomic variables (mortgage interest rate, money supply, rent index, consumption index, and average salary), and monthly stock market sector indices (63 indices between 2009 and 2020). In addition, we acquired data on the costs of construction materials—such as cement, metals, wood, plastics, paints, and electrical and mechanical equipment—and construction labor.

Mean normalization was conducted on the aforementioned diverse data to standardize the ranges of the independent variables x . The means μ of the independent variables were calculated using Equation 13.

$$x' = \frac{x - \mu}{\max(x) - \min(x)} \quad (13)$$

E. Statistical Analysis

The clustering results were statistically analyzed. IBM SPSS Statistics was used to perform descriptive statistical analysis, normality tests, analysis of variance (ANOVA), and linear regression. The descriptive statistics were the mean, maximum, minimum, variance, standard deviation, coefficient of skewness, and coefficient of kurtosis [43]. The skewness and kurtosis coefficients were employed to determine the state of deviation from normal distributions. ANOVA is appropriate when the response variables have continuous distributions and the conditions are discrete, either by nature or by design [44]. The ANOVA results were evaluated on the basis of R^2 , adjusted R^2 , sum of squares, mean square, F value, and significance level (α) [45]. R^2 represents the proportion of variance in the dependent variable that the independent variable

can explain; a higher R^2 value thus indicates higher explanatory power. The adjusted R^2 value is a modified version of R^2 that considers additional independent variables that tend to bias R^2 results. The mean square is the estimated variance used to determine the accuracy of the regression model, and the sum of squares is the sum of the squares of the distances between each data point and the line of best fit. The F value is used to compare the amount of variance explained by the model with the amount of error or unexplained variance to determine the test's significance.

The results of the linear regression included the standardized regression coefficient β , T value, tolerance, variance inflation factor (VIF), Pearson correlation coefficient, and P value [46]. The standardized regression coefficient is used to compare the explanatory power of the independent variables and calculate the slope of the regression equation. Moreover, the T value indicates whether a significant difference exists between the variables, with a higher T value indicating stronger evidence. The VIF is used to assess the independence of the independent variables in multiple linear regressions, and a small VIF value indicates a high degree of independence. The Pearson correlation coefficient is employed to test the linear relationship between two variables. Regression analysis was conducted to determine which relationships in the developed model had statistically significant coefficients and P values.

III. RESULTS

Identifying the factors that directly or indirectly influence the number of real estate transactions is of interest to many research institutions; however, identifying such factors is a complex task. To identify the aforementioned factors, we collected and analyzed 144 monthly data from 2009 to 2020 on all industrial market stock indices from the TWSE that are related to real estate. The number of features is the value of the variable for each month from 2009 to 2020, and the 144 elements are used for dimensionality reduction using an autoencoder. We also collected data on socioeconomic indicators and sociodemographic factors from various sources, including the Ministry of the Interior, national statistics, the Ministry of Finance, and open government data. In total, data on 98 monthly indices were collected, and these data were organized, classified, and standardized.

A. Analysis of DBM-PSO clustering performance

The genetic algorithm (GA) [47], PSO [37], differential evolution (DE) [48], arithmetic optimization algorithm (AOA) [49], whale optimization algorithm (WOA) [50], gray wolf optimization (GWO) [51], and biogeography-based optimization (BBO) [52] were selected to evaluate the clustering performance of DBM-PSO. All algorithms used the same two control parameters, i.e., the maximum number of iterations is set to 100, and the total number of individual is set to 50. The other parameters of the optimization algorithms used the default parameter according to the aforementioned

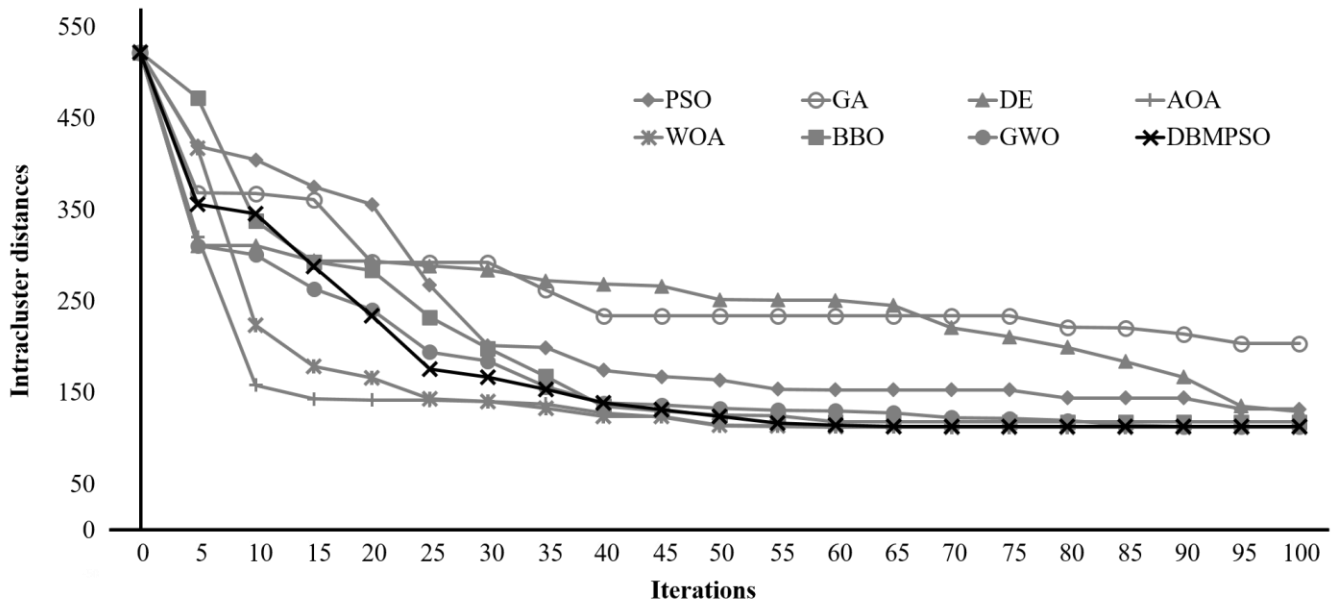


Figure 3. Comparisons of intracluster distances of clustering result on eight clustering algorithms.

literatures. The intra-cluster distance comparisons of the clustering results for the eight clustering algorithms are shown in Figure 3. The DBM-PSO outperformed the PSO, GA, DE and BBO for clustering results. The closest intracluster distances were obtained by DBM-PSO (111.87), AOA (111.04), WOA (111.01), and GWO (111.79). The small difference of the intra-cluster distance is due to the fact that the best solutions among the four clustering algorithms have a small difference between the centroid positions of the clusters. Furthermore, the AOA, WOA and GWO showed that DBM-PSO achieve the optimal clustering performance.

B. Analysis of DMB-PSO clustering solution

Detailed information on the collected data is provided in the following text. The DBM-PSO clustering analysis of these indices indicated the existence of six clusters of variables. In six clusters, the three clusters contained 15 (denoted as cluster 1), 38 (cluster 2), and 13 (cluster 3) variables that showed the significant result related with real estate. The variables in cluster 1 exhibited strong correlations with the number of real estate transactions, and we further explored these variables. Furthermore, the variables in cluster 2 exhibited significant associations with the average unit price of real estate transactions. Finally, the variables in cluster 3 had significant relationships with the building materials and construction index and thus influenced real estate costs.

The following sections describe the results of the aforementioned three clusters in detail. IBM SPSS Statistics was used to validate the data clustering results. This study aimed to create a deep learning approach for analyzing and predicting real-estate-related trends and determined how socioeconomic and sociodemographic factors influence real estate trends. The focus of this study was on identifying the

variables that significantly influence the number of real estate transactions, the average unit price of real estate transactions, and the building materials and construction index.

Number of real estate transaction

The variables that were discovered to be significantly related to the number of real estate transactions were as follows:

- (1) Taiwan monthly population growth rate,
- (2) Monthly population growth rate from 15 to 64 years old,
- (3) Construction salary index,
- (4) Mortgage interest rate,
- (5) Construction equipment rental index,
- (6) Construction asphalt index,
- (7) Shipping and transportation index,
- (8) Construction brick tile index,
- (9) Optoelectronic return index,
- (10) Wood products index,
- (11) Optoelectronic index,
- (12) Civil engineering labor index,
- (13) Shipping and transportation return index,
- (14) Construction paint coating index,
- (15) Construction engineering salary index.

The dimensionality of high-dimensional data was reduced using autoencoders before clustering was performed. The average values and DBM-PSO clustering results of the aforementioned 15 variables are presented in Figure 4. The optimal sum of intracluster distances was obtained in cluster 1. In Figure 4, the height of each bar represents the average value of a variable, and the positive and negative numbers indicate the highest and lowest values, respectively, of the variable. Regression analysis is a reliable method for identifying the

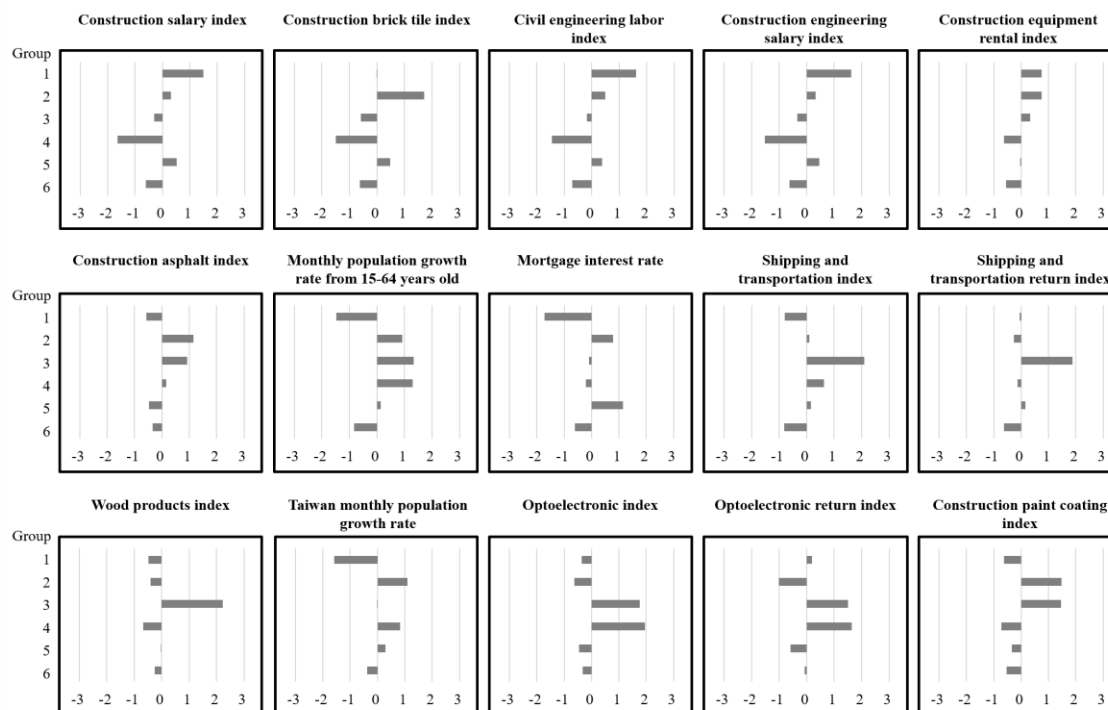


Figure 4. Average value of characters in clustering results related to number of real estate transactions.

influence of an independent variable on a dependent variable. Figure 5 shows the results of the regression analysis of the effects of the aforementioned 15 independent variables on the number of real estate transactions (dependent variable).

Table 1 presents the ANOVA results for the number of real estate transactions. Moreover, Table S1 details the descriptive statistics of the aforementioned 15 indicators. The R^2 value for the correlations between the 15 independent variables and the number of real estate transactions was 0.9933, which indicated the high power of the 15 indicators to explain the number of real estate transactions. We analyzed the descriptive statistics of the 15 variables to identify the most important factors influencing the number of real estate transactions. In addition to population growth, the salary, transportation, and construction indices exhibited strong correlations with the number of real estate transactions.

Average unit price of real estate transactions

We attempted to identify the factors that influence home prices by analyzing indices that are strongly correlated with the average unit price of real estate transactions. The unit price of a home is a critical consideration for prospective buyers, and understanding the complex set of factors that influence home prices is essential for predicting economic trends and consumer behavior. Home prices can significantly affect people's willingness to buy real estate and can reflect the state of the economy. Through clustering analysis, we identified the following 37 indices that significantly influenced the average unit price of real estate transactions:

- (1) Paper and pulp index,
- (2) Construction cement products index,
- (3) Plastic return index,
- (4) Electrical and cable return index,
- (5) Iron and steel return index,
- (6) Chemical return index,
- (7) Oil gas and electricity index,
- (8) Oil gas and electricity return index,
- (9) Communications and internet index,
- (10) Number of first-time real estate registrations,
- (11) Taiwan monthly population growth rate,
- (12) Monthly population growth rate from 0-14 years old,
- (13) Monthly population growth rate over 65 years old,
- (14) Cement return index,
- (15) Food index,

Table 1. Analysis of variance (ANOVA) for number of real estate transactions.

	Sum of Squares	d.f.	Mean square	F	Sig	R^2
Regression	143.032	15	9.535	1260.375	.000*	0.993
Residual	0.968	128	0.008			
TOTAL	144.000	143				

Dependent variable: number of real estate transactions

*Independent variable: Taiwan monthly population growth rate, monthly population growth rate from 15 to 64 years old, Construction Salary index, mortgage interest rate, Construction Equipment rental index, Construction asphalt index, Shipping and Transportation index, Construction Brick Tile index, Optoelectronic return index, Wood products index, Optoelectronic index, Civil Engineering labor index, Shipping and Transportation return index, Construction Paint coating index, Construction Engineering Salary index.

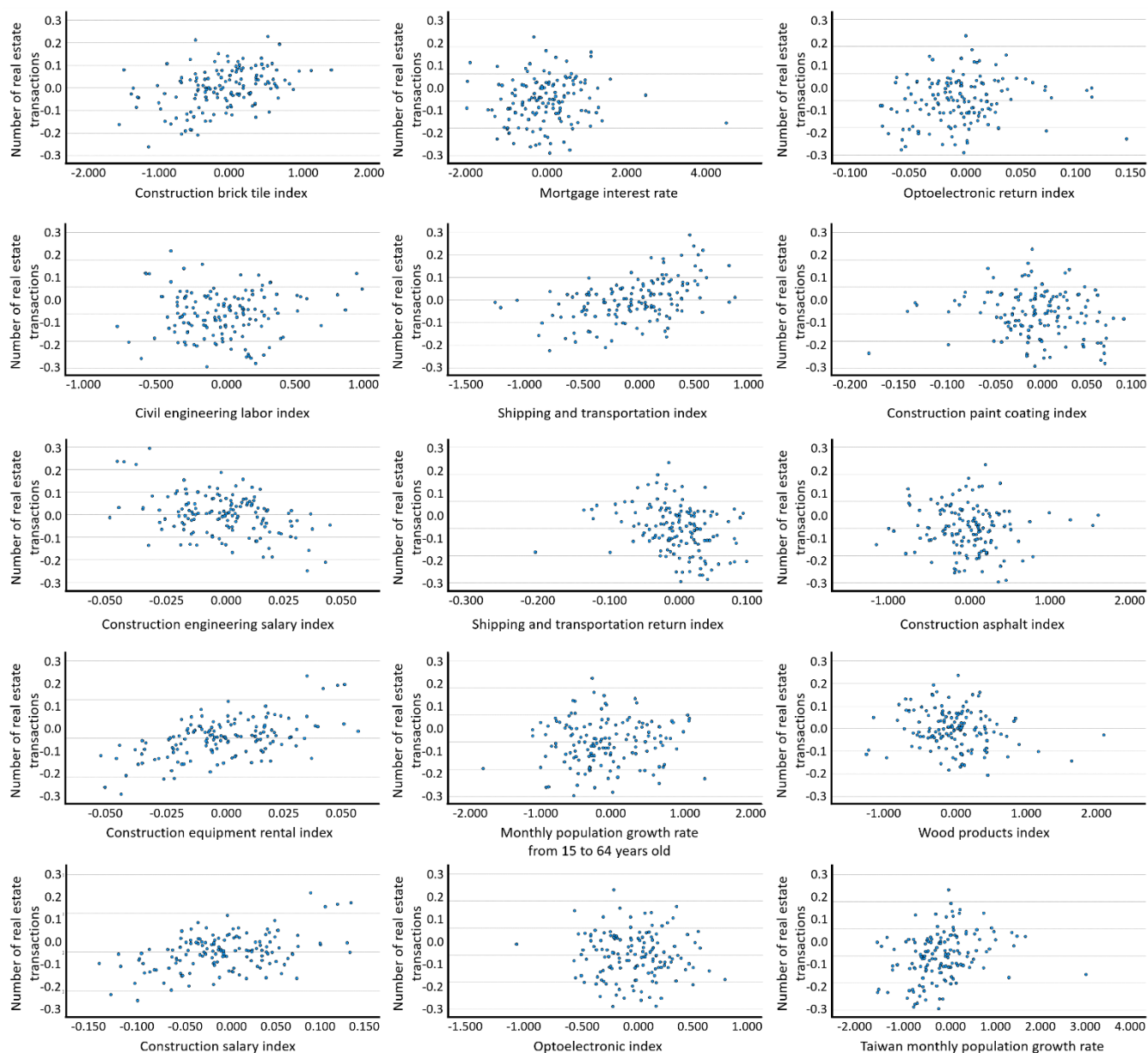


Figure 5. Regression analysis of all indexes in cluster to number of real estate transactions.

- (16) Food return index,
- (17) Textile return index,
- (18) Electric machinery index,
- (19) Electric machinery return index,
- (20) Paper and pulp return index,
- (21) Building material and construction return index,
- (22) Finance and insurance index,
- (23) Finance and insurance return index,
- (24) Trading and consumers goods return index,
- (25) Others return index,
- (26) Semiconductor index,
- (27) Semiconductor return index,
- (28) Computer and peripheral return equipment,
- (29) Electronic parts components index,
- (30) Electronic parts components return index,

- (31) Electronic products distribution return index,
- (32) Information service index,
- (33) Information service return index,
- (34) Other electronic indexes,
- (35) Other electronic return indexes,
- (36) Electronic return index,
- (37) Electrical index.

These 37 indices were extracted using techniques such as data standardization, high-dimensional model reduction by using autoencoders, and data clustering. The DBM-PSO results for these indices are illustrated in Figure 6. The clustering of groups provided an optimal sum of intracluster distances within all clusters. Each bar in Figure 6 indicates the average value of one of the aforementioned 37 variables, and the positive and negative

numbers indicate the highest and lowest values of the variable, respectively.

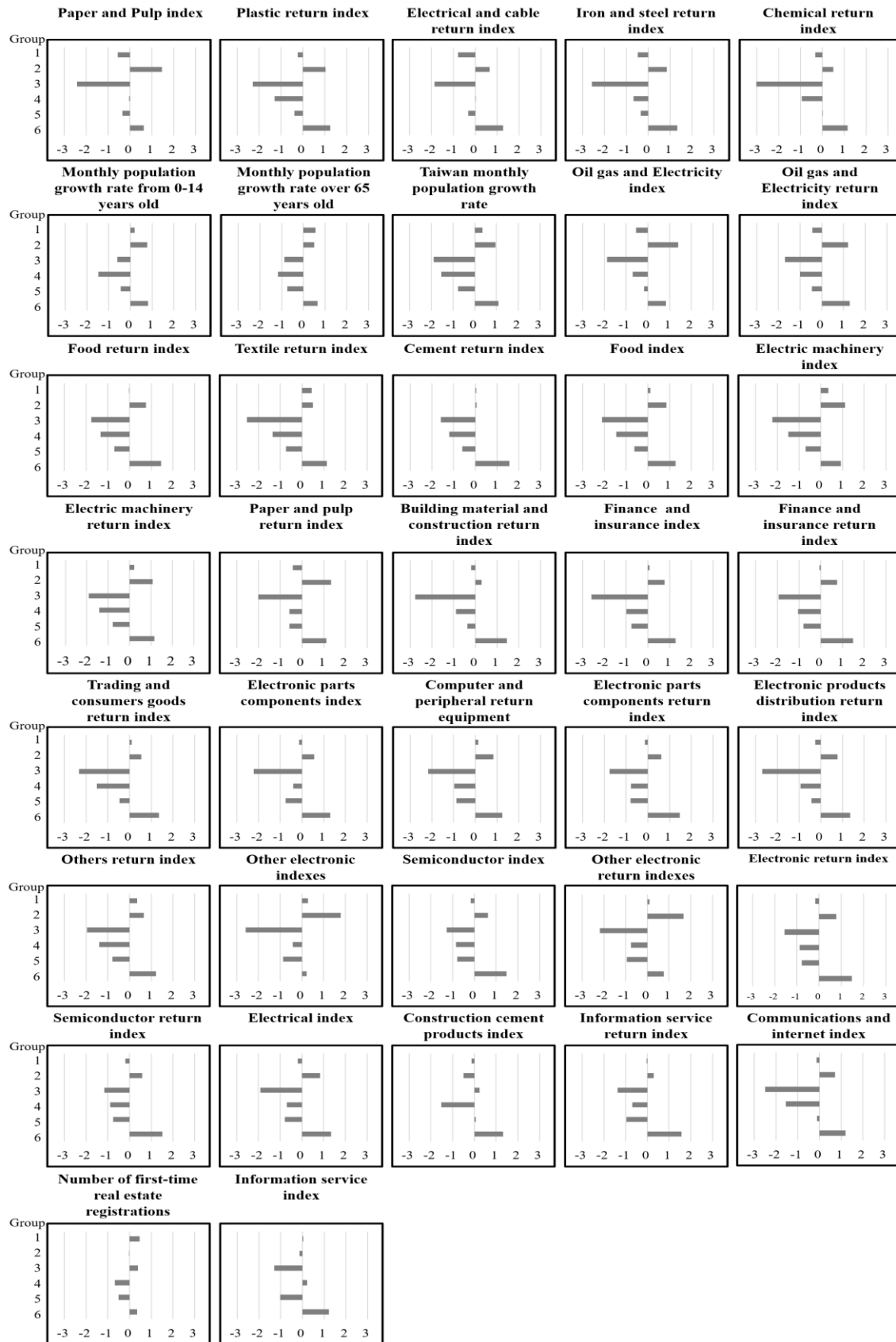


Figure 6. Average value of characters in clustering results related to average unit price of real estate transactions.

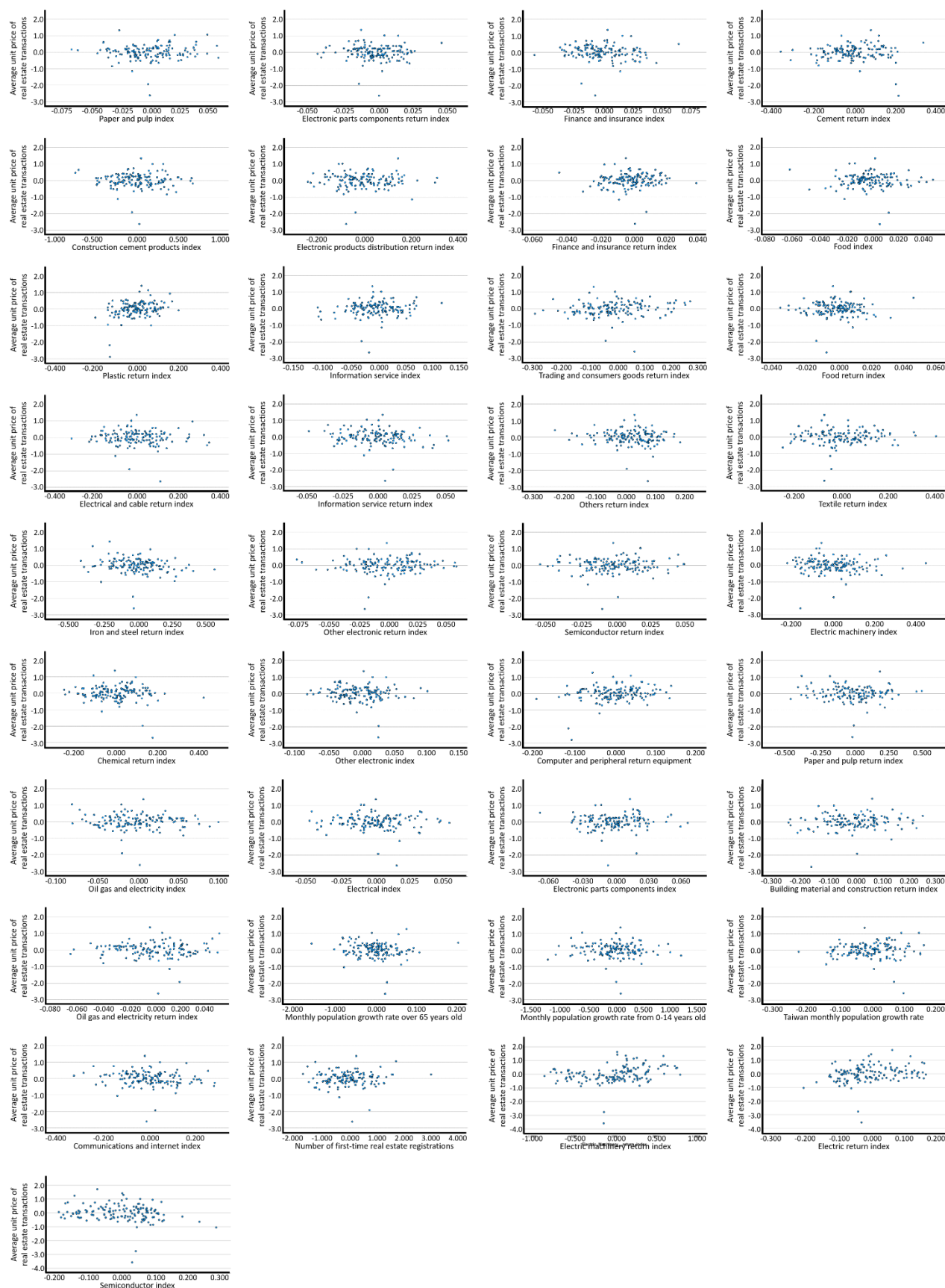


Figure 7. Regression analysis of all indexes in cluster to average unit price of real estate transactions.

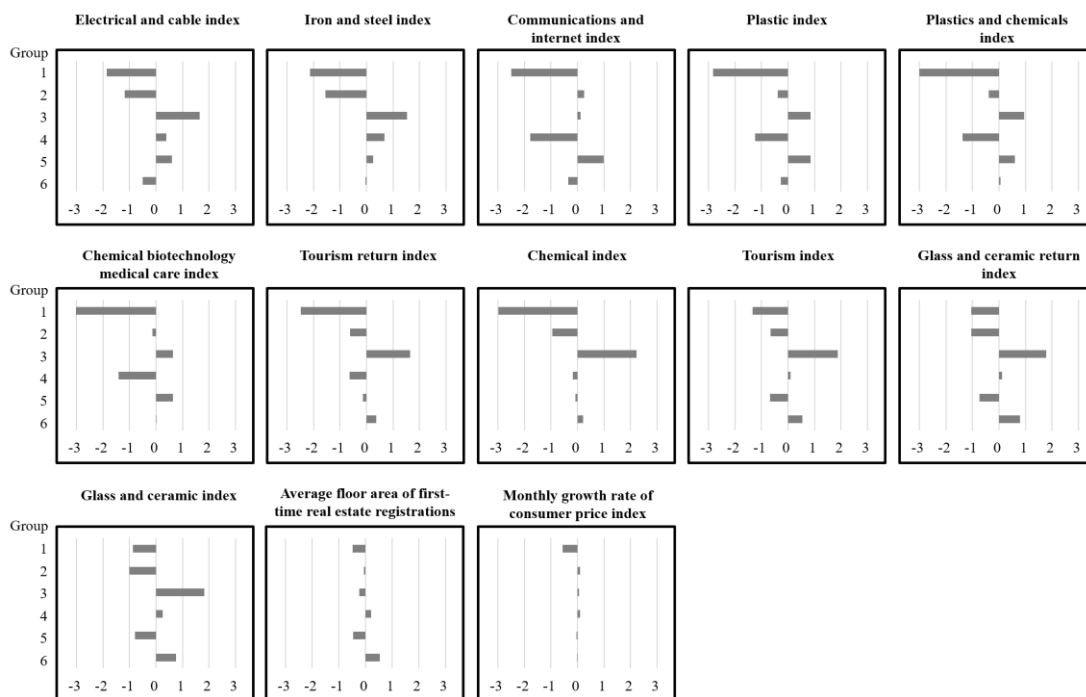


Figure 8. Average value of characters in clustering results related to building material and construction index.

Table S2 presents the indicators that influenced the average unit price of real estate transactions. IBM SPSS Statistics was used to perform ANOVA, normality tests, and linear regression. The ANOVA results for the aforementioned 37 variables indicated that R^2 was 0.7960, which indicated that the 37 independent variables had high power to explain the average unit price of real estate transactions (Table 2). In addition to population growth, transportation, and construction indices exhibited significant correlations with the average unit price of real estate transactions. Figure 7 illustrates the regression analysis results of the aforementioned 37 independent variables on the average unit price of real estate transactions.

Building material and construction index

We identified 13 indices that exhibited strong correlations with the building materials and construction index. Accurately predicting construction and building material costs is crucial not only for developers but also for real estate investors. By knowing construction costs, investors can anticipate builders' pricing strategies, devise investment plans, and determine when to buy and sell properties. Furthermore, knowledge of construction costs can assist homebuyers in identifying the optimal time to purchase a home. The 13 indices that exhibited strong correlations with the building materials and construction index were as follows:

- (1) Average floor area of first time real estate registrations,
- (2) Iron and steel index,
- (3) Monthly growth rate of consumer price index,
- (4) Communications and internet index,

- (5) Tourism return index,
- (6) Glass and ceramic return index,
- (7) Electrical and cable index,
- (8) Chemical biotechnology medical care index,
- (9) Plastic index,
- (10) Chemical index,
- (11) Tourism index,
- (12) Plastics and chemicals index,
- (13) Glass and ceramic index.

Table 2. Analysis of variance (ANOVA) for average unit price of real estate transactions.

	Sum of Squares	d.f.	Mean square	F	Sig	R^2
Regression	114.620	34	3.371	12.509	.000*	0.796
Residual	29.380	109	0.270			
TOTAL	144.000	143				

Dependent variable: average unit price of real estate transactions

*Independent variable: Electrical index, Number of firsttime real estate registrations, Construction Cement products index, Oil Gas and Electricity index, monthly population growth rate over 65 years old, monthly population growth rate from 014 years old, Other electronic indexes2, Paper and Pulp index, Information Service index, Chemical return index, Textile return index, Iron and Steel return index, Communications and Internet index, Electric Machinery index, Finance and Insurance index, Food index, Electrical and Cable return index, Cement returnindex, Trading and Consumers Goods return index, Building Material and Construction return index, Electronic Products Distribution return index, Electronic Parts Components index, Plastic return index, Computer and Peripheral return Equipment, others return index, Taiwan monthly population growth rate, Information Service return index, Oil Gas and Electricity return index, Other electronic return indexes, Semiconductor return index, Paper and Pulp return index, Food return index, Electronic Parts Components return index, Finance and Insurance return index

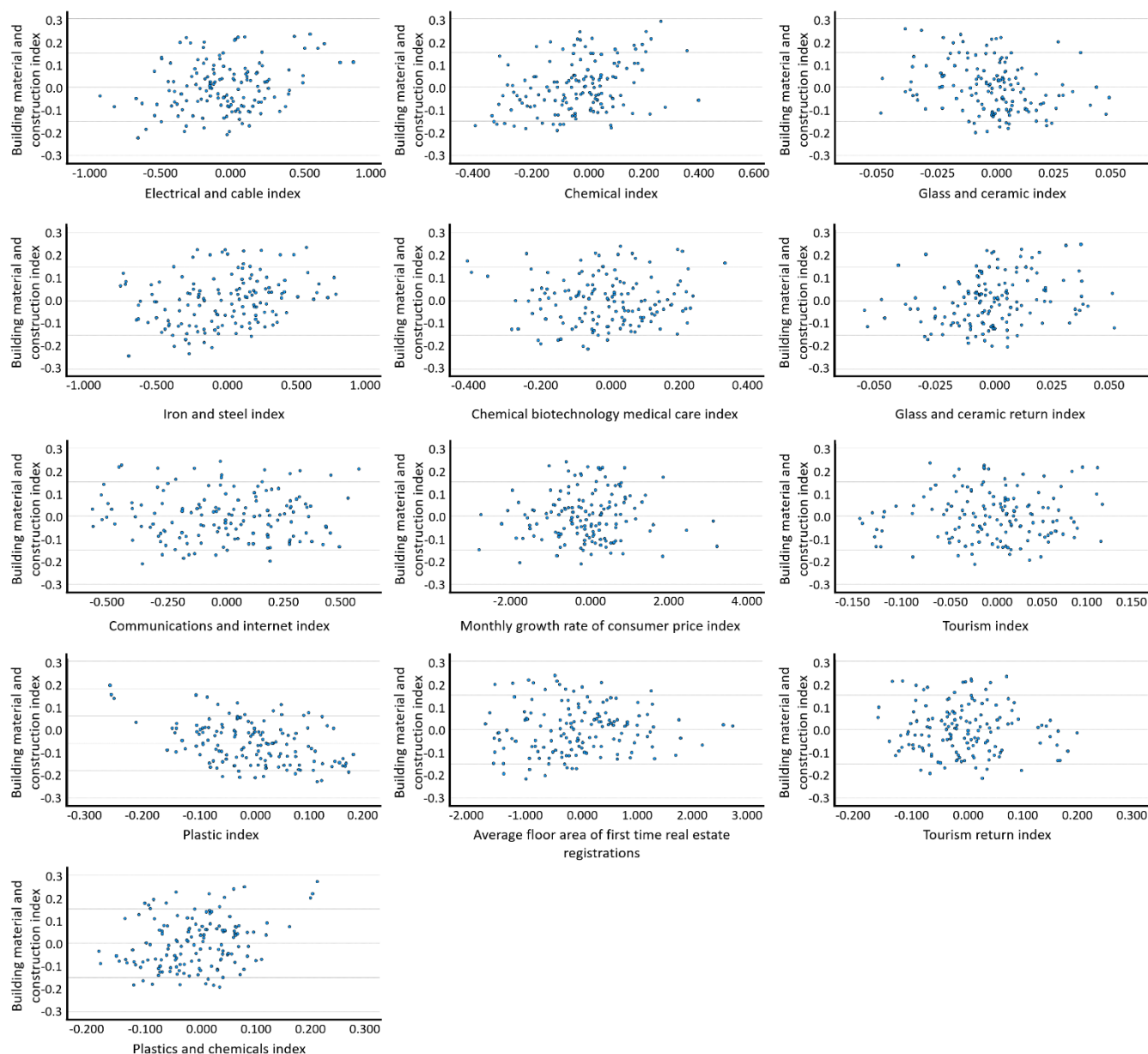


Figure 9. Regression analysis of all indexes in cluster to building material and construction index.

Figure 8 shows the DBM-PSO results for the clustering of the aforementioned 13 indices. The number of clustering groups was determined in accordance with the optimal sum of intracluster distances that resulted in the most efficient clustering. Each bar in Figure 8 represents the average value of the clustered samples for each index that affects the building materials and construction index, and the positive and negative numbers indicate the highest and lowest values of the variable, respectively.

Descriptive statistical analysis, normality tests, ANOVA, and linear regression were performed using IBM SPSS Statistics. After data clustering, the descriptive statistics were analyzed (Table S3). The high average R^2 value of the aforementioned 13 independent variables (0.8822) in the

ANOVA indicated that these variables had high power to explain the building material and construction index (Table 3). The regression analysis results for the aforementioned 13 variables are shown in Figure 9.

IV. DISCUSSION

Due to advancements in deep learning, deep clustering has received considerable attention. Deep clustering involves combining a clustering method with the automatic learning representation of high-dimensional, high-semantic data by using a deep neural network (DNN). The clustering task has been redefined by the deep adaptive clustering of Chang et al. [53] and its extensions [54], in which clustering is considered a binary pairwise classification problem for determining the

correlation between low-dimensional DNN-learned representations. Other methods simultaneously learn feature representations by using a DNN and cluster assignments and by using the k -means algorithm. For example, Yang et al. [55] introduced the DeepCluster method for jointly learning representations by using a convolutional neural network. In this method, learning representations are clustered using the k -means algorithm. Ge et al. [56] proposed dual adversarial autoencoder, which is a clustering method that maximizes mutual information within the framework of an adversarial autoencoder. However, the aforementioned methods have the drawbacks of distorted representations and poor clustering performance because they treat representation learning and clustering as separate processes. Therefore, in the proposed approach, the deep-clustering problem is formulated as minimization of the distance from a joint distribution of observed data, learning representations, and soft cluster assignments within an autoencoder framework. This approach enables the learning of meaningful representations and the capture of structural information from observation data. In conclusion, our method contributes to the field of deep clustering by allowing the learning of soft cluster assignments within an autoencoder framework, the imposition of constraints on the input data and representations to prevent bias in the soft cluster assignments, and the use of the DBM-PSO algorithm to balance the learning of informative representations and clustering. Our experimental results indicate the excellent performance of the proposed clustering method and the effectiveness of learning representations for predicting and analyzing real estate data.

Clustering is a valuable technique for segmenting economic and functional data and has been widely applied in various studies [57]. The k -means algorithm has been widely used in credit rating and real estate investment research but has limitations in clustering analysis [17]. To improve clustering accuracy, we adopted a group-based optimization method that involves improved PSO. The advantage of group-based optimization is that it can be used to obtain efficient recurrent structures or the best overall cluster patterns [58]. Moreover, a segment-based approach was used for investigating the effects of individual segments on real estate by evaluating the similarities of features across data sets. The DBM-PSO clustering algorithm was employed to formulate the soft-constrained clustering problem as a single optimization problem instead of examining the effect of a single feature on real estate in a hard-constrained problem [59]. In this approach, the behavioral effects of other features are treated as entirely a part of the model, which results in poor noise recognition capability. The DBM provided three high-frequency regions (with chaotic values of 0.0, 0.5, and 1.0) that varied over time, thereby effectively balancing particle search behavior between exploration and exploitation. The DBM-PSO algorithm outperforms other clustering algorithms because of its superior ability to explore and exploit in the search process. The DBM-PSO algorithm has a more accurate update equation but not

Table 3. Analysis of variance (ANOVA) for building material and construction index.

	Sum of Squares	d.f.	Mean square	F	Sig	R ²
Regression	127.044	13	9.773	74.923	.000*	0.882
Residual	16.957	130	0.130			
TOTAL	144.000	143				

Dependent variable: building material and construction index

*Independent variable: Average floor area of firsttime real estate registrations, Iron and Steel index, monthly growth rate of Consumer price index, Communications and Internet index, Tourism return index, Glass and Ceramic return index, Electrical and Cable index, Chemical Biotechnology Medical Care index, Plastic index, Chemical index, Tourism index, Plastics and Chemicals index, Glass and Ceramic index.

higher complexity than does the PSO algorithm. The DBM-PSO clustering algorithm defines K centers for K clusters. The Euclidean distance is used for evaluation of the intracluster distance, and a shorter intracluster distance indicates a better clustering solution. The best clustering solution ($gbest$) is obtained in the last iteration. Each sample in the data set is assigned to a cluster in accordance with the shortest distance between the sample vector and the cluster. The resulting segmented tags provide insight into the characteristics of each group, which can help in real estate marketing and analysis of marketing strategies.

In this study, the feasibility of using deep learning for predicting trends in real estate data was investigated. All the collected data were preprocessed, standardized, and clustered using high-dimensional models with autoencoders. Three data clusters were extracted, with the first cluster representing factors affecting the number of real estate transactions, the second cluster representing factors affecting the average unit price of real estate transactions, and the third cluster representing factors affecting the building material and construction index. These three key indicators are fundamental in shaping real estate trends.

The results obtained for the first cluster serve as a reference for governmental policy intervention. For example, in Taiwan, the number of real estate transactions surged from 2010 to 2014, which caused the government to implement policies for controlling the cost of real estate between 2015 and 2016 (Figure 1). However, most studies focusing on macroeconomics and examining the overall dynamics of the real estate market while ignoring other factors [60]. Taiwan's housing prices and housing transactions during 2011–2015, when the loan-to-value ratio was reduced and a transfer tax along with a property tax were collected [61]. The aforementioned increase in transaction volume was driven by the monthly population growth rate of 15–64-year-olds because of their strong demand for home ownership. Bank loans are found to be much more significant than interest rates in predicting the movements of both asset prices. This suggests that Taiwan's asset price fluctuations support the theory that emphasizes the importance of balance sheet position and collateral value to credit-constrained firms [62]. Roundtrip transaction costs for buying properties are high,

above 10% in several countries in Asia. High transaction costs and unaffordable housing combined with weak protection of property rights are seen to be conducive to the creation of urban slums [63]. Moreover, correlations existed between building materials, salaries, and transaction volume. The collected data indicated that the increase in construction-related costs (salaries, materials, tools, transportation, and paint), rather than in housing prices, had driven the number of real estate transactions, which is a new discovery.

The average unit price of real estate transactions affects various factors, including consumer perceptions of future real estate market conditions. The key indicators in cluster 2 included the monthly growth rate of the population aged over 65 years, the building materials and construction yield index, and some key stock market indicators. This study makes a major contribution to general real estate forecasting, which is typically based on macroeconomic forecasts. This study provides individual forecasts for the unit price of real estate transactions and for transaction volume while controlling for other relevant variables. Our findings are noteworthy and suggest that a growth in the number of 15–64-year-old people drives real estate transaction volume, whereas a growth in the number of those aged >65 years drives unit price growth. In addition, increases in construction-related costs and the building materials and construction yield index lead to an increase in transaction volume. Finally, mortgage interest rates lead to an increase in transaction volume, whereas the stock market earnings index drives unit price growth.

This study found that the building materials and construction index exhibited positive correlations with the average floor area of first-time property registrations and the monthly growth rate of the consumer price index (CPI). When the CPI increases, developers tend to construct larger properties, which leads to an increase in construction costs. Apart from indicators related to construction materials, the increase in construction costs is driven by several other factors, such as the CPI, real estate market trends, and indicators related to construction materials.

Creating a transaction-based real estate price index through statistical analysis is challenging because of the insufficient information available in the real estate industry, where full disclosure is often not preferred by decision makers [64]. Therefore, the use of deep-learning approaches to predict real estate trends is a viable alternative. Hudson et al. indicated that machine learning and big data applications are increasingly being used in the development of smart and sustainable cities [65]. To achieve sustainable social development in smart cities, citizen-centric data governance mechanisms and data-driven value creation are essential [66]. In the future, smart city technologies will incorporate Internet of Things devices and sociotechnical networked systems that align organizational and institutional processes across urban governance [67].

V. Conclusion

In this study, cluster analysis and deep-learning techniques were used to identify the key factors influencing Taiwan's sustained housing price increases over the past decade. Data collection, data standardization, autoencoder-based dimensionality reduction, and data clustering were performed in this study, and three data clusters were obtained. The first cluster contained independent variables that influence the number of real estate transactions; the second cluster contained independent variables that influence the average unit price of real estate transactions; and the third cluster comprised independent variable indices that affect the building materials and construction index when controlling for relevant variables. The framework proposed in this paper provides a new mechanism for approaching real-estate-related forecasting, thereby enabling researchers and governments to focus on specific aspects of real estate development without being influenced by other related factors. Future research should focus on expanding data collection efforts, exploring different clustering algorithms for subgroup analyses, and optimizing parameter settings. Our finding might be useful for governmental regulation of the relationship between the rental market and the real estate market; thus, considering people's housing equity when formulating real-estate-related laws and regulations is essential. Although this study only focused on Taiwan, the proposed framework can be applied globally to analyze housing markets in different regions.

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